Finite Rotational Symmetry and Certified Error Budgets for a Voronoi Pivotal Amplitude

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M423

October 9, 2026

Abstract

The assigned manuscript derives a Voronoi color-pivotal amplitude from a scalar Cardy premise and substantial companion and annular-transfer inputs. We isolate its signed four-interface module. Averaging over any cyclic rotation group of order at least three eliminates the only frequency obstruction to the stronger spectral gap; full isotropy is unnecessary for this step. The source's own Lipschitz decay then improves its projected forcing estimate, removing a logarithmic loss and attaining the separated-end exponent 9/4. Conditional on the source's discrete exploration and domination machinery, we obtain a three-rate transfer budget for finite-rotation-invariant even messages, angular defects, and color defects. We also optimize the stated near/far insertion balance and give explicit residual, integral, and logarithmic-amplitude enclosures when their constants are actually certified. These results preserve the distinction between an intensity insertion derivative and the color Russo derivative. Physical-metric and omitted-space proofs from the original Ram corpus guide the finite-certificate boundaries; classical Fourier projection, semigroup estimates, and error balancing are credited. No numerical value of the amplitude, unconditional Cardy theorem, or independent global universality proof is supplied. Contribution assessment: C3/5, a source-specific finite-symmetry extension with endpoint signed decay and a quantitative defect budget, conditional on the original continuum and discrete transfer inputs.

Contents

1 Model, derivatives, and normalization **2**
1.1 Two distinct differentiation identities 2
1.2 The calibration is not a numerical constant 3

2 Source dependencies and literature **3**

3 The actual chamber and its symmetry modes **4**

4 Finite rotations remove the slow obstruction **5**

5 Endpoint decay and residual budgets **6**
5.1 What an approximate symmetry costs 6

6 Rough shapes and the discrete message interface **7**
6.1 Even messages need only finite rotational invariance 8

7 From insertion bounds to an amplitude enclosure **9**
7.1 Optimizing the two-size balance 9
7.2 Finite integral and infinite-tail budgets 11
7.3 Planar integrability and the symmetry defect 11

8 Physical representation and original Ram support **12**

9 Gains, costs, and limitations **12**

10 Evidence scope **13**

1 Model, derivatives, and normalization

The source [1] uses a full-plane Poisson process of intensity ε^{-2}, with independently fair black/white cell colors. Crossings use closed Voronoi cells in an unwired quadrilateral. A nucleus is color-pivotal if changing its color alone changes the crossing. Nuclei outside the quadrilateral count when their cells meet it. All expectations are annealed over both geometry and colors.

The source assumes convergence to Cardy's value for each bounded Jordan quadrilateral with four distinct marked boundary points. It claims

$$\varepsilon^{3/4}\mathbb{E}N_\varepsilon([0,1]^2) \longrightarrow c_V \in (0,\infty).$$

This is a conditional amplitude claim, not a conclusion obtained by differentiating a qualitative limit. We retain that status throughout.

At unit intensity let $Q_0 = [-1,1]\times[-1/2,1/2]$, let X_R be its scaled left–right crossing indicator, and define

$$H_R(\eta) = X_R(\eta\cup\{0^{\mathrm{b}}\}) - X_R(\eta\cup\{0^{\mathrm{w}}\}), \qquad f(R) = \mathbb{E}H_R(\eta).$$

The inserted central point is forced black and white in the same geometry; monotonicity gives $H_R \in \{0,1\}$. For an independent fair insertion at $y \neq 0$, put

$$K_R(y) = \mathbb{E}[H_R(\eta\cup\{y^{\mathrm{f}}\}) - H_R(\eta)]. \tag{1}$$

This signed kernel need not be nonnegative and is not a Poisson-pivotal probability. Its integral uses physical Lebesgue area dy.

1.1 Two distinct differentiation identities

For a fixed finite geometry, the Bernoulli product-rule calculation gives

$$\frac{d}{dp}\mathbb{P}_p(X_R = 1 \mid \eta) = \sum_{v:\,\mathrm{cell}(v)\cap RQ_0\neq\varnothing} \mathbb{P}_p(v \text{ is color-pivotal} \mid \eta).$$

Indeed condition on all colors except v; the derivative in that coordinate is the black-forced minus white-forced expectation. Sum the coordinate derivatives. The number of cells meeting a fixed bounded quad has finite mean: localization reduces to points in a bounded neighborhood except for exponentially unlikely large cells, whose Poisson first moments are integrable. This geometry-only

bound also dominates the derivative uniformly in p. Thus annealing gives the color Russo identity. It does not give the derivative in intensity.

For the latter, if η_t has intensity t, Euclidean scaling gives

$$G_R(t) := \mathbb{E}H_R(\eta_t) = f(\sqrt{t}\,R), \qquad \frac{R}{2}f'(R) = \int_{\mathbb{R}^2} K_R(y)\,dy. \tag{2}$$

Here the factor $1/2$ comes from differentiating $\sqrt{t}$. In a bounded region D, the Poisson series gives the score

$$\frac{d}{dt}\mathbb{E}F(\eta_t \cap D) = \mathbb{E}\left[F(\eta_t \cap D)\left(\frac{N_t(D)}{t} - |D|\right)\right], \qquad \mathbb{E}\left(\frac{N_t(D)}{t} - |D|\right)^2 = \frac{|D|}{t}.$$

The equivalent add-one formula integrates the fair-color insertion. The source justifies the full-plane limit by nested localization screens: a difference supported on a failure of probability $O(e^{-c4^n})$ has score derivative $O(2^n e^{-c'4^n})$. Both series are uniformly summable on compact intensity intervals. For fixed R, distant insertion differences are exponentially dominated. This proves the C^1 identity, but the resulting constants depend on R; the later *normalized* tail estimate is a different and substantially stronger input.

1.2 The calibration is not a numerical constant

Write $\lambda = 5/4$, $S_\varepsilon = f(1/\varepsilon)$, $q_\varepsilon = (\mathbb{E}N_\varepsilon(Q_0))^{-1}$, and $m_\varepsilon = \varepsilon^{-2}$. The imported companion calibration is

$$m_\varepsilon q_\varepsilon S_\varepsilon \to c > 0, \qquad q_\varepsilon \mathbb{E}N_\varepsilon([0,1]^2) \to B > 0.$$

If $R^\lambda f(R) \to a > 0$, the exact normalization is

$$c_V = \frac{Ba}{c}. \tag{3}$$

The rectangle Q_0 and unit square are not interchangeable; B accounts for that comparison. Even a rate for $R^\lambda f(R)$ does not give a square-count rate without rates for both calibration limits.

2 Source dependencies and literature

All 11 cached texts, including the 960-line transfer proof and the 518-line amplitude proof, were literally read in bounded displays. The following is a dependency map, not an independent global audit.

Module	Mechanism and retained obligation
Inputs	Companion arm bounds, screened locality, good graph sequences, Jordan tests, stopped explorations, calibration, and exponent without amplitude.
Annuli	Two-cap identity; prime-end banks kept distinct; boundary-motion control; disk-patch gluing; admissible stopped annuli and collisions.
Spectrum	Annulus SLE locality and modulus clock; singular chamber diffusion; ground-state transform; noncollision, smoothing and symmetry gaps.
Shape	Capacity and conformal-radius end maps; round-collar reflection and an angular error with a logarithm, uniform over rough boundaries.
Transfer	Independent core before conditioning, first four-tip frontiers, six-arm tails, rank-one factorization, symmetry cancellation, failed-screen resets and projective wedge contraction.
Amplitude	Intensity differentiation; signed tail with the factor $f(R)$; side/corner probes; two-size determinant; dyadic logarithmic derivative and final companion calibration.

Several distinctions are indispensable. The message norm is an essential supremum over successful output seams of *total variation*, with the sum-of-absolute-masses convention. Successful screens are indicators, not conditioning followed by renormalization. Auxiliary core variables are independent of the original seams; this yields uniform operator errors without a scaling limit conditional on every atypical history. The source's rank-one estimate alone does not give the even signed decay. Its wedge estimate is in a projective tensor norm and is used only through successful cuts. Failed cuts are treated by the last-failure partition, not ignored.

The primary literature refresh on October 9, 2026 distinguishes three statuses. Du–Gao–Li–Zhuang [2], in their primary preprint Theorems 1.1–1.3, obtain sharp arm asymptotics for triangular-lattice critical site percolation; the displayed alternating four-arm result has a positive amplitude and an $o(1)$ relative error. This is not a Voronoi amplitude theorem. Vanneuville's primary abstract [3] states annealed Voronoi scaling relations, quasi-multiplicativity and pivotal estimates; its full proof was not audited here. The current publisher repository overview [4] also lists a separate Cardy manuscript and a quenched near-critical companion. Those are publisher claims, not independent verification of their proofs. We do not silently replace this report's Cardy premise with those claims.

3 The actual chamber and its symmetry modes

For one labeled tuple use

$$\mathcal{C} = \{w_1 < w_2 < w_3 < w_4 < w_1 + 2\pi\}/(2\pi\mathbb{Z}\mathbf{1}).$$

Only common full turns are quotiented out; cyclic relabeling is a separate operation. On $\mathcal{X} = \mathcal{C} \times \mathcal{C}$, set

$$J(w) = \prod_{i<j} \sin\frac{w_j - w_i}{2}, \quad h(x,y) = J(x)^{1/3} J(y)^{1/3}, \quad d\mu = Z^{-1} J(x)^{4/3} J(y)^{4/3}\, dx\, dy.$$

The factors are positive in the open chamber and vanish at collisions. All Hilbert norms in this section use μ, not unweighted angle measure or a sampling grid. The source proves that the ground-state transform of its transmission equation is

$$W = e^{\lambda p}(T^+ - T^-)/h, \qquad \partial_p W = DW + B_p \cdot \nabla W + V_p W, \quad D = \frac{3}{8}(\Delta + \nabla \log\mu \cdot \nabla), \tag{4}$$

where $\|B_p\|_{C^1} + \|V_p\|_\infty = O(e^{-p})$. The notation $\nabla \log\mu$ means the gradient of its density. Tangency of B_p removes the apparent collision singularities in $V_p = B_p \cdot \nabla \log h$.

We import the source's noncollision and smoothing proof, including the semigroup $P_t = e^{tD}$, its symmetry on $L^2(\mu)$, and

$$\|P_1\|_{L^2(\mu)\to L^\infty} < \infty.$$

The constant is not numerically evaluated. The source also proves, uniformly on the chamber for sufficiently large p,

$$\|W(p)\|_\infty + \operatorname{Lip} W(p) \le Ae^{-3p/4}. \tag{5}$$

These are continuum, whole-space inputs. A finite Ritz matrix does not prove them.

Write $\theta_w = \frac{1}{4}\sum_i w_i \pmod{2\pi}$ and $r_w = w - \theta_w \mathbf{1}$, with compatible real lifts. The relative coordinate belongs to the convex zero-sum alcove. The measure splits into the uniform center circle and relative law. Each center contributes $(3/32)\partial_{\theta_w}^2$. For a zero-sum relative vector v,

$$\nabla^2(-\log J^{4/3})[v,v] = \frac{1}{3}\sum_{i<j} \csc^2\frac{r_i - r_j}{2}(v_i - v_j)^2 \ge \frac{4}{3}|v|^2.$$

The source applies the classical Brascamp–Lieb inequality to obtain relative Poincare gap 1/2, with Dirichlet factor 3/8. We use that inequality with its actual domain and measure.

Let $\mathcal{H}_- \subset L^2(\mu)$ consist of simultaneous-rotation invariant functions that change sign under one cyclic relabeling of either tuple. The signed W belongs to this space. Simultaneous rotation restricts the center frequencies to $(n, -n)$. A relabeling increases one center by $\pi/2$; therefore the partial relative mean of such a mode vanishes unless $n \equiv 2 \pmod 4$. The source's energy accounting gives the modewise lower bound

$$\kappa(n) = \frac{3n^2}{16} + \mathbf{1}_{\{n \not\equiv 2 \pmod 4\}}. \tag{6}$$

The indicator is the sum of two 1/2 relative gaps, one for each tuple. This bound, rather than a guessed finite spectrum, is the key to the extension below.

4 Finite rotations remove the slow obstruction

For a probability measure ω on the angular circle define

$$(\mathcal{Q}_\omega F)(x, y) = \int F(x, y + \phi\mathbf{1})\, d\omega(\phi), \qquad \widehat{\omega}(n) = \int e^{in\phi}\, d\omega(\phi).$$

It is a contraction on $L^2(\mu)$ and L^∞, and commutes with D. It need not be a projection. For the uniform measure on the N-th roots of unity, write $\mathcal{Q}_N$; this is an orthogonal projection onto modes $n \in N\mathbb{Z}$.

Theorem 4.1 (A finite angular annihilator)**.** *On $\mathcal{H}_-$, the image of $\mathcal{Q}_\omega$ has Dirichlet lower bound* 1 *whenever $\widehat{\omega}(2) = 0$. Consequently, for $F \in \mathcal{H}_-$,*

$$\|P_t \mathcal{Q}_\omega F\|_2 \le e^{-t} \|\mathcal{Q}_\omega F\|_2.$$

In particular this holds for every $\mathcal{Q}_N$, $N \ge 3$. It also holds for the two-point filter

$$\mathcal{Q}_* = \tfrac{1}{2}(I + \mathcal{R}_{\pi/2}), \qquad (\mathcal{R}_\phi F)(x, y) = F(x, y + \phi\mathbf{1}). \tag{7}$$

The two-point filter is not a projection. Two equally spaced half-turn samples, namely $\mathcal{Q}_2$, do not give this conclusion.

Proof. The only integers with the lower bound $\kappa(n) < 1$ in (6) are $n = 2, -2$. At $n = 0$ the bound is 1; at $n \not\equiv 2 \pmod 4$ it is at least 1; and the remaining nonzero integers have $|n| \ge 6$, with center energy at least 27/4. The angular filter multiplies mode n by $\widehat{\omega}(-n)$. Real nonnegative weights give conjugate multipliers at opposite frequencies, so the hypothesis kills both exceptional modes. Orthogonality of Fourier modes and the source's form inequality prove the lower bound; the symmetric semigroup gives the exponential estimate by the energy inequality.

For $\mathcal{Q}_N$, the multiplier is zero unless N divides n. No $N \ge 3$ divides 2. For $\mathcal{Q}_*$, the multiplier at $n = 2$ is $(1 + e^{i\pi})/2 = 0$. The two half-turn samples instead have multiplier 1 at that mode. □

Example 4.2 (The exact slow mode)**.** The real function

$$F_2(x, y) = \cos(2(\theta_x - \theta_y))$$

is simultaneously rotation invariant and cyclic-odd in each tuple. It is constant in the relative variables and satisfies $DF_2 = -3F_2/4$. Thus $W(p) = e^{-3p/4}F_2$ solves the unperturbed transformed equation. It is fixed by $\mathcal{Q}_2$, whereas $\mathcal{Q}_3$, $\mathcal{Q}_4$, and $\mathcal{Q}_*$ kill it. Multiplication by $e^{-5p/4}h$ gives decay exactly e^{-2p}. Its radial area integral is at the logarithmically divergent threshold. This is a continuum generator example, not a claim that arbitrary transmission probabilities can be prescribed to equal this function.

The two-point rule is minimal among positive finite angular filters annihilating frequency 2: one point has a unit-modulus multiplier. For two points, a convex combination of two unit complex numbers is zero only when their weights are $1/2, 1/2$ and the numbers are antipodal. Their angles must therefore differ by $\pi/2$ modulo π. This is an exact algebraic rule, not a quadrature convergence assertion.

5 Endpoint decay and residual budgets

The source first bounds the forcing in (4) by $O(e^{-p})$, obtaining $(1+p)e^{-p}$ after full rotation projection. Its subsequent Lipschitz estimate (5) permits a second use of the same equation. No new probabilistic input is needed for this improvement.

Theorem 5.1 (No resonant logarithm)**.** *Assume the source's whole-chamber evolution, symmetry, smoothing and* (5)*. For every probability filter with $\widehat{\omega}(2) = 0$,*

$$\|\mathcal{Q}_\omega W(p)\|_\infty \le Ce^{-p}, \qquad \|\mathcal{Q}_\omega (T^+ - T^-)(p)\|_\infty \le Ce^{-9p/4}. \tag{8}$$

The constants can be chosen uniformly over these filters.

Proof. Put $F_p = B_p \cdot \nabla W(p) + V_p W(p)$. The actual Lipschitz bound, not merely a supremum bound on W, gives

$$\|F_p\|_\infty \le F_0 e^{-7p/4}.$$

The forcing remains in $\mathcal{H}_-$. Projection is applied to this complete forcing; we do not commute $B_p \cdot \nabla + V_p$ with independent rotations.

Let $S = \|P_1\|_{2\to\infty}$, and put $C_s = e\max(1, S)$. For every bounded function Z in the spectral subspace with gap at least 1,

$$\|P_t Z\|_\infty \le C_s e^{-t}\|Z\|_\infty \quad (t \ge 0). \tag{9}$$

For $t \ge 1$, use $P_1 P_{t-1}$, the L^2 gap and $\|Z\|_2 \le \|Z\|_\infty$; for $t \le 1$, use the Markov supremum contraction and $e^{-t} \ge e^{-1}$. Fix a starting modulus p_0 in the range of the imported estimates. Duhamel and Theorem 4.1 imply

$$\|\mathcal{Q}_\omega W(p)\|_\infty \le C_s e^{-(p-p_0)}\|W(p_0)\|_\infty + C_s F_0 \int_{p_0}^{p} e^{-(p-s)} e^{-7s/4}\, ds.$$

The integral equals

$$\frac{4}{3} e^{-p}(e^{-3p_0/4} - e^{-3p/4}),$$

which proves the first estimate. The function h is bounded and invariant under independent center rotations, so it commutes with the filter. Multiplication by $e^{-\lambda p}h$ proves the second. □

5.1 What an approximate symmetry costs

Let $\epsilon_2 = |\widehat{\omega}(2)|$ for any probability filter. Split W into its $n = \pm 2$ modes and the remaining modes. Each individual Fourier coefficient has supremum norm at most $\|W\|_\infty$; the complementary projection has supremum norm at most three. Repeating the preceding proof on the complement gives

$$\|\mathcal{Q}_\omega (T^+ - T^-)(p)\|_\infty \le Ce^{-9p/4} + 2A\|h\|_\infty \epsilon_2 e^{-2p}. \tag{10}$$

The constant in the first term is uniform over probability filters. Signed weights of total variation L instead incur the corresponding factor L; unbounded signed weights are not covered for free.

For two equal weights at angles $0, \pi/2 + \delta$, $\epsilon_2 = |\sin\delta| \le |\delta|$. If the first weight is $1/2 + \epsilon$ and the second is $1/2 - \epsilon$, with $|\epsilon| \le 1/2$, the combined error satisfies

$$\epsilon_2 \le 2|\epsilon| + |\delta|.$$

This follows by comparing with the exactly weighted rule at the same angles and using the triangle inequality. A fixed small error produces an e^{-2p} term; small is not the same as integrable over all scales.

Proposition 5.2 (Whole-forcing, area-weighted certificate)**.** *In the gap-one subspace, suppose a bounded mild solution $Z(t)$, $t \ge 0$, has forcing $q(t)$ with*

$$Z(t) = P_t Z(0) + \int_0^t P_{t-s} q(s)\, ds, \quad z_0 = \|Z(0)\|_\infty, \quad r(s) = \|q(s)\|_\infty.$$

Then

$$\int_0^\infty e^{3t/4} \|Z(t)\|_\infty \, dt \le 4C_s \left(z_0 + \int_0^\infty e^{3s/4} r(s)\, ds \right). \tag{11}$$

All integrals may initially have value $+\infty$.

Proof. Use (9), multiply by $e^{3t/4}$, and apply Tonelli to the nonnegative upper bound. The initial integral is $4z_0$. For fixed s, the forcing integral is $\int_s^\infty e^{3t/4} e^{-(t-s)} dt = 4e^{3s/4}$. □

The weight $3/4 = 2 - \lambda$ is forced by planar area in logarithmic radius. A residual bounded by $\varepsilon e^{-\gamma t}$ contributes at most $4C_s\varepsilon/(\gamma - 3/4)$ when $\gamma > 3/4$. A uniform small residual or the endpoint decay $e^{-3t/4}$ does not give a finite bound by this criterion. To apply it to a finite approximation, q must include the full projected forcing and all discarded and arithmetic residuals. A small compressed residual does not establish the hypothesis.

6 Rough shapes and the discrete message interface

Take the source's separated Jordan ends $K_i \subset U_o$, with

$$K_i \subset \{|z| \le a\}, \quad \operatorname{diam} K_i \ge c_* a, \quad \{|z| < b\} \subset U_o, \quad \min_{\partial U_o} |z| \le C_* b.$$

The marked accesses remain distinct, but no uniform smoothness or mark separation is assumed. Let $\rho = a/b$, $q = \log(1/\rho)$; the constants below depend on c_*, C_*, not on the shapes. The source's end-map comparison gives modulus error $O(\rho)$ and angular error $\eta = C(1+q)\rho$, relative to the single-end capacity and conformal-radius coordinates.

Corollary 6.1 (Finite-filter separated-end endpoint)**.** *For a probability filter ω, rotate one physical end, with all its marks and prescribed colors, through its sampled angles. Then*

$$\left| \int (T_\phi^+ - T_\phi^-)\, d\omega(\phi) \right| \le C\rho^{9/4} + C|\widehat{\omega}(2)|\rho^2. \tag{12}$$

In particular the exact $9/4$ exponent holds for every $N \ge 3$ cyclic average and for the two-quarter-separated-point filter.

Proof. The source's signed shape comparison, using the spatial Lipschitz, time 1/2-Holder, and h 1/3-Holder estimates, costs

$$C\rho^2\{\rho^{1/2} + \eta + \eta^{1/3}\}.$$

Dividing this by $\rho^{9/4}$ gives terms bounded by constants times $\rho^{1/4}$, $(1+q)\rho^{3/4}$, and $(1+q)^{1/3}\rho^{1/12}$. They are bounded for small ρ; the last is not a logarithmic loss at exponent 9/4. The single-end capacities are invariant under rotation and its angles shift exactly by ϕ. Apply (10) at $p' = \log(l_o/l_i) = q + O(1)$, and add the uniform shape error. The argument for the other end uses simultaneous rotation invariance. □

This reaches the endpoint excluded by the source's $(1+q)\rho^{9/4}$ estimate. It does not improve its unsigned rank-one exponent or assert a discrete mesh-convergence rate.

6.1 Even messages need only finite rotational invariance

Let $\mathcal{B}_r$ be the source Banach space of seam-measurable, zero-mass signed connection-state measures, vanishing on seam failure, with norm

$$\|v\|_r = \operatorname*{ess\,sup}_{\Sigma_r} \|v_{\Sigma_r}\|_{\mathrm{TV}}.$$

Rotations act on both geometry and states; color reversal acts on colors and pairings. These are commuting isometries. Write

$$\mathcal{E} = \tfrac{1}{2}(I + \mathcal{F}), \qquad \mathcal{A}_N = \frac{1}{N}\sum_{j=0}^{N-1} \mathcal{R}_{2\pi j/N}.$$

These are contractions and commuting projections in the Banach space, not orthogonal Hilbert projections. Their meaning differs from $\mathcal{Q}_N$ on chamber functions, although the symmetries agree.

Theorem 6.2 (Three-rate message budget)**.** *Retain the source's exploration, independent-core domination, admissibility, six-arm first-reduction tails, and continuum replacement inputs. Fix $N \geq 3$. There are $C, \delta > 0$ such that, for every sufficiently large fixed dyadic ratio L, a threshold $r_0(L)$ exists with*

$$\|M_{r,Lr}|_{\mathrm{even},C_N}\| \leq CL^{-2-\delta}, \qquad \|M_{r,Lr}|_{\mathrm{even}}\| \leq CL^{-2} \quad (r \geq r_0(L)).$$

Together with the source's general $CL^{-\lambda}$ bound, this gives

$$\begin{aligned}\|M_{r,Lr}v\|_{Lr} \leq C(&L^{-2-\delta}\|\mathcal{A}_N\mathcal{E}v\|_r + L^{-2}\|(I-\mathcal{A}_N)\mathcal{E}v\|_r \\ &+ L^{-\lambda}\|(I-\mathcal{E})v\|_r).\end{aligned} \tag{13}$$

All conclusions retain the essential supremum over output seams. The constants and threshold are not numerically evaluated.

Proof. At the first four-tip frontier, the pushed message is $\alpha(H_i)(\delta_B - \delta_W)$. Evenness makes $\alpha(\mathcal{F}H_i) = -\alpha(H_i)$, because the reference cap difference changes sign. Color symmetry of the unweighted end law therefore replaces transmission by one half of $T^+ - T^-$, exactly as in the source. Without angular invariance, the unsigned bound on this signed difference is $C(a_j/b_k)^2$, from (5) and the shape error. With C_N invariance, average the N rotated histories: the coefficient and law are unchanged, and Corollary 6.1 gives $C(a_j/b_k)^{9/4}$.

Here is the quantitative summation needed for the operator norm. For first-reduction levels j, k, the source proves unweighted history mass at most $C2^{-(2+\tau)(j+k)}$, for some $\tau > 0$, and $a_j/b_k \asymp L^{-1}2^{j+k}$. For any

$$2 < q_* < \min(9/4, 2+\tau)$$

the used-pair sum is at most

$$CL^{-q_*} \sum_{j,k\geq 1} 2^{-(2+\tau-q_*)(j+k)} \leq C'L^{-q_*}.$$

For the even-only bound take $q_* = 2$, where the same series converges. The unused pairs cost $C(1+\log L)^2 L^{-2-\tau}$. Absorb that factor with a smaller $\delta < q_* - 2$ in the first bound, or directly into CL^{-2} in the second.

The signed coefficients are pointwise bounded by their input TV norm, and the output cap difference has TV at most two. Thus the source's independent-core domination lemma controls each continuum replacement uniformly over all original seams; no conditioning on a typical history is introduced. Fix L first and then take $r_0(L)$ large enough to absorb the finitely many qualitative errors. This is exactly the source's order of choices. Finally decompose v into the three displayed components and apply the triangle inequality. □

For unit input norm, an angular defect of size d_a and a color defect of size d_c contribute at scales $L^{-2}d_a$ and $L^{-5/4}d_c$. To match $L^{-2-\delta}$, sufficient budgets are

$$d_a \leq L^{-\delta}, \qquad d_c \leq L^{-3/4-\delta}. \tag{14}$$

These are sufficient one-block budgets, not necessary conditions or a claim that a numerical message obeys them. Exact group averaging can enforce the symmetries, but modifies the message; the removed component must be charged if the original observable is wanted.

The operators commute with the exact finite group actions. Therefore an even C_N-invariant message remains in that class under composition, and the source's fixed-ratio iteration applies after absorbing the per-block constant. The last-failed-screen argument also preserves these finite symmetries when the screens and reset laws do. The actual fair insertion is already fully isotropic, so its original bound is retained; the extension accommodates a larger class of signed inputs, rather than creating a new pivotal observable.

7 From insertion bounds to an amplitude enclosure

This section keeps the remaining stochastic inputs explicit. It supplies quantitative consequences of those inputs, not a second proof of their geometric or probabilistic validity.

7.1 Optimizing the two-size balance

Assume that $f > 0$ is C^1 for large R, satisfies (2), and $f(R) = R^{-\lambda+o(1)}$. Assume the source estimates, for fixed $\xi, \nu' > 0$,

$$\int_{|y|>m} |K_R(y)|\, dy \leq Cf(R)(m^{-\xi} + R^{-\xi}), \tag{15}$$

$$|K_R(y)f(R') - K_{R'}(y)f(R)| \leq C(m/R)^{2\lambda+\nu'} + Ce^{-cm^2}. \tag{16}$$

The first holds for $m_0 \leq m \leq R/100$; the second for almost every $|y| \leq m$ and $100m \leq R \leq R' \leq 2R$, above fixed thresholds. Every constant is uniform over these quantified choices. The source proves

them using signed messages, side/corner arms, and projective wedge contraction, not merely bulk unsigned four-arm estimates.

Theorem 7.1 (Optimized conditional central rate)**.** *Put $D_* = 2 + 2\lambda + \nu'$. For every*

$$0 < \alpha < \alpha_* := \frac{\xi\nu'}{D_* + \xi} \tag{17}$$

the preceding assumptions imply

$$\left| \frac{Rf'(R)}{f(R)} + \lambda \right| \le C_\alpha R^{-\alpha}, \qquad R^\lambda f(R) = a(1 + O_\alpha(R^{-\alpha}))$$

for some $a \in (0, \infty)$. Thresholds and constants may depend on α, and need not be effective.

Proof. Choose an exponent loss $\eta > 0$ so small that

$$\beta_* = \frac{\xi(\nu' - 2\eta)}{D_* + \xi} > \alpha, \qquad 0 < 2\eta < \nu',$$

and set $\zeta = (\nu' - 2\eta)/(D_* + \xi)$, $m = R^\zeta$. Then $0 < \zeta < 1$, so the scale restrictions hold for large R. The exponent assumption gives $f(s) \ge s^{-\lambda-\eta}$ for all sufficiently large s. Divide (16) by $f(R)f(R')$ and integrate over the disk of area πm^2. The power term is bounded by

$$CR^{-\nu'+2\eta+\zeta D_*} = CR^{-\beta_*}.$$

The exponential error is smaller than any inverse power after multiplication by $R^{2\lambda+2\eta}m^2$. The two normalized outer integrals cost $CR^{-\zeta\xi} = CR^{-\beta_*}$. Hence

$$\sup_{R \le R' \le 2R} |g(R') - g(R)| \le C_\alpha R^{-\alpha}, \qquad g(R) = Rf'(R)/f(R). \tag{18}$$

The weakening from β_* to α harmlessly absorbs the remaining error terms.

Summing (18) along $R, 2R, 4R, \dots$ gives a limit γ, with $|g(R) - \gamma| \le C_\alpha R^{-\alpha}/(1 - 2^{-\alpha})$. The comparison between each pair of consecutive dyadic scales gives the same limit for all real R, not only a subsequence. Since $\log f(R) - \log f(R_0) = \int_{R_0}^R g(s)\, ds/s$, the exponent assumption identifies $\gamma = -\lambda$. Now the derivative of $\log(R^\lambda f(R))$ is absolutely integrable at infinity. Its finite real limit gives $a > 0$, and the tail integral is $O(R^{-\alpha})$, proving the relative error statement. □

The source chooses $\zeta_0 = \nu'/(4D_*)$, $\eta_0 = \nu'/8$, obtaining $\min(\nu'/2, \zeta_0\xi)$. The optimized value balances the two affine exponents instead of reserving a fixed quarter-budget. This is a classical optimization applied to the actual source inequalities. It does not establish new values of ν' or ξ. For a manufactured admissible tuple $\lambda = 5/4, \nu' = 1/2, \xi = 1/4$,

$$D_* = 5, \qquad \alpha_* = \frac{1}{42}, \qquad \min(\nu'/2, \zeta_0\xi) = \frac{1}{160}.$$

Taking $\eta = 1/100$ gives $\zeta = 16/175$ and $\beta_* = 4/175$, so for example $\alpha = 1/50$ is allowed. These are exact arithmetic examples, not measured Voronoi exponents.

7.2 Finite integral and infinite-tail budgets

Suppose at one scale R that

$$|\widehat{f} - f(R)| \le e_f < \widehat{f}, \qquad \left|\widehat{I} - \int_{|y|\le m} K_R(y)\,dy\right| \le e_I.$$

If $T_R(m)$ is a certified upper bound on the normalized absolute tail in (15), direct division gives

$$\left|g(R) - \frac{2\widehat{I}}{\widehat{f}}\right| \le 2\left[\frac{e_I}{\widehat{f} - e_f} + \frac{|\widehat{I}|e_f}{\widehat{f}(\widehat{f} - e_f)} + T_R(m)\right]. \tag{19}$$

To verify it, subtract the two quotients before taking absolute values; use $f(R) \ge \widehat{f} - e_f > 0$. The integral error must include geometry, expectation, quadrature, and omitted-cell errors appropriate to the chosen implementation. Agreement between two cutoffs or two quadrature rules is not such an enclosure.

An infinite-scale amplitude interval requires more than (19) at one or finitely many scales. If the all-future estimate (18) holds with a *known* constant C for every $R \ge R_0$, the proof of Theorem 7.1 gives

$$R^\lambda f(R) e^{-E_R} \le a \le R^\lambda f(R) e^{E_R}, \qquad E_R = \frac{CR^{-\alpha}}{\alpha(1 - 2^{-\alpha})}. \tag{20}$$

One may replace $f(R)$ by its positive lower and upper enclosures in the two endpoints. For $c_V = Ba/c$, positive enclosures for the *limiting* B, c, or certified errors to those limits, are additionally needed. The source's qualitative calibration and unspecified starting scales do not themselves provide those numbers.

There is also a useful non-power version. If $\sup_{R\le s\le 2R} |g(s) - g(R)| \le \omega(R)$ and $\sum_{j\ge 0} \omega(2^j R) < \infty$, then

$$|g(R) + \lambda| \le \Omega(R) := \sum_{j\ge 0} \omega(2^j R),$$

provided the same preliminary exponent identifies the limit. A positive finite amplitude follows if additionally $\int_{R_0}^\infty \Omega(s)\,ds/s < \infty$. The two requirements are distinct. For $g(R) = -\lambda + 1/\log R$, $R > e$, dyadic oscillation is $O((\log R)^{-2})$, but $f(R) = CR^{-\lambda}\log R$; the normalized amplitude diverges. Summable oscillation alone gives a derivative limit, not necessarily an amplitude.

7.3 Planar integrability and the symmetry defect

In logarithmic scale $p = \log(r/a)$, the physical factor $e^{-\lambda p}$ and planar area $r\,dr$ leave the weight $e^{(2-\lambda)p} = e^{3p/4}$ in Proposition 5.2. Likewise the shape estimate gives, for a fixed inner scale a, the illustrative bounds

$$\int_M^\infty r\,(a/r)^{9/4}\,dr = 4a^{9/4}M^{-1/4}, \qquad \int_M^\infty r\,(a/r)^2 \epsilon_2(r)\,dr = a^2 \int_M^\infty \epsilon_2(r)\,\frac{dr}{r}.$$

An angular defect with a finite logarithmic integral is therefore compatible with this tail bound; a fixed defect is not. These integrals explain the source's threshold and the cost of an inexact filter. They do not on their own bound the normalized pivotal kernel: that step still requires the source's central and enclosing arm factors, side/corner probes, and localization losses that retain the small factor $f(R)$.

8 Physical representation and original Ram support

The freshly read original TR248 passages [5] prove a full finite-model leakage identity, physical mass scaling, and rigorous entry/Gram perturbation budgets. The original book proofs [6, 7] give the precise distinction between a compression and a full residual. We use these as methodological support, not as a substitute for annulus SLE or for the source's Banach-space domination lemma.

For example, in the present Hilbert space $H = L^2(\mu)$, let $J_0 : \mathbb{C}^k \to H$ be injective synthesis by form/domain-admissible trial functions. The physical Gram and compression of $-D$ are

$$G = J_0^* J_0, \quad K = J_0^*(-D)J_0, \quad B = G^{-1}K, \quad P = J_0 G^{-1} J_0^*.$$

For domain vectors the exact residual identity is

$$\|(-D-\zeta)J_0 c\|_H^2 = \|(B-\zeta)c\|_G^2 + \|(I-P)(-D)J_0 c\|_H^2. \tag{21}$$

It follows by orthogonal projection, not by a convergence experiment. A certified finite matrix gap only controls the trial space. In our argument the complementary-space bound comes instead from (6) and the analytic Poincare inequality. The collision-degenerate density must be retained in G; omitting it changes the norm. Grid evaluation also does not certify a whole-chamber Lipschitz bound or the forcing required by (11).

There is a second, different norm in the problem: $\mathcal{B}_r$ has seam-essential-supremum TV norm, not this Hilbert norm. No Gram equivalence is asserted between them. The passage from a continuum supremum estimate to the discrete TV operator estimate is the independently specified core-domination argument. Confusing these spaces would lose the uniformity over arbitrary signed messages and atypical output seams that the source carefully proves.

9 Gains, costs, and limitations

The notable local gain is a weaker angular symmetry hypothesis with a sharp account of the slow obstruction. A finite positive filter can annihilate the exact offending harmonic; no continuum angular quadrature is intrinsically needed for this module. The endpoint 9/4 follows from the source's already available regularity, and the three-rate bound says how imperfect evenness and angular symmetry differ in cost. This is not simply a relabeling of a finite matrix.

The price is that these conclusions rely on whole-space spectral, smoothing, and regularity statements. They do not make the source's qualitative mesh threshold computable. Nor do they replace its six-arm/frontier topology, independent-core comparison, wedge estimate, or companion calibration. A finite filter does not prove the existence of an isotropic Poisson scaling limit, and a symmetric sample is not a proof that its unseen remainder is symmetric. No run-time improvement or measured numerical amplitude is claimed.

Feasible next steps are narrowly defined: certify explicit smoothing and coefficient constants on a truncated chamber with controlled collision collars; build exact angular filter weights and retain their frequency-two error; and supply genuine all-future insertion-tail and calibration moduli. Any implementation should expose its physical measure, source of every tail bound, and whether a tested message or the actual full message is being certified. A broader model with only finite rotational symmetry would additionally require new discrete convergence and geometry proofs; this report does not supply them.

No mathematical correction to the source is asserted on this scope. The logarithmic bound and conservative exponent choice in the source are valid bounds; the present results sharpen or extend them under their retained premises. Applicability remains L3. C3 reflects the finite-symmetry

transfer extension and its explicit defect separation. It is not C4: the global Cardy-to-pivotal chain has not been independently proved and no stochastic input or amplitude constant has been computed.

10 Evidence scope

The owned read ledger records hashes captured before fresh displays, the exact untruncated intervals, explicit unread complements, and unchanged post-read endpoints. All 11 public texts were read in full, 3,299 lines. The original TR248 was read at 263–306, 398–457, and 1151–1248; the two original book chapters were read at 1–210 and 1–242. Their included figures and cited antecedent reports were not thereby read. The core5 locator was hashed but not freshly read or used as proof. No whole-corpus or complete external-literature audit is claimed.

Owned exact tests check the finite angular annihilator, modewise gap conditions, group projections, three-rate arithmetic, forcing integrals, near/far balancing and quotient enclosures. Counts of parameter cases and assertions are recorded separately in the executable receipt: 810 cases and 2,177 exact assertions passed. They are not percolation simulations, verification of SLE convergence, or a replay of foreign checkers. Parent checks are excluded.

References

[1] OpenAI, *A Pivotal Amplitude for Voronoi Percolation from Cardy's Formula*, October 5, 2026. Pinned public manuscript source. All cached texts read; captured October 8 byte identity is provenance, not proof verification or live-branch status.

[2] H. Du, Y. Gao, X. Li and Z. Zhuang, *Sharp asymptotics for arm probabilities in critical planar percolation*, arXiv:2205.15901v1 (2022); Commun. Math. Phys. 405 (2024), Article 182. `https://arxiv.org/abs/2205.15901`. Primary preprint introduction and main statements read, not full proof.

[3] H. Vanneuville, *Annealed scaling relations for Voronoi percolation*, Electron. J. Probab. 24 (2019), Paper 39. `https://arxiv.org/abs/1806.08452`. Primary abstract and version metadata checked.

[4] OpenAI, public mathematics release overview, family 224, checked October 9, 2026. Publisher overview. Companion claims only; their original proofs were not read here.

[5] M. Subbarao, original author-supplied private TR248 revision, `TR-rev-1-old-tr248.tex`, lines 263–306, 398–457, 1151–1248. Exact leakage, physical metrics, and finite certificate proofs. Confidential local source; not uploaded or searched online.

[6] M. Subbarao, original author-supplied Ram Calculus Volume 21, `ch01d-physical-metrics.tex`, lines 1–210. Physical Gram, full residual, relative metric and finite-measurement proofs.

[7] M. Subbarao, original author-supplied Ram Calculus Volume 21, `ch01f-spectral-limits.tex`, lines 1–242. Spectral pollution, strong-versus-norm convergence, domain-aware inversion and dual residual proofs. Confidential local source.